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Chapter 3

4. Consider a swapping system in which memory consists of the following hole sizes in memory order: 10 MB, 4 MB, 20 MB, 18 MB, 7 MB, 9 MB, 12 MB, and 15 MB. Which hole is taken for successive segment requests of
   (a) 12 MB
   (b) 10 MB
   (c) 9 MB
   for first fit? Now repeat the question for best fit, worst fit, and next fit.
6. For each of the following decimal virtual addresses, compute the virtual page number and offset for a 4-KB page and for an 8 KB page: 20000, 32768, 60000.
7. Using the page table of Fig. 3-9, give the physical address corresponding to each of the following virtual addresses:
   (a) 20
   (b) 4100
   (c) 8300
14. A machine has a 32-bit address space and an 8-KB page. The page table is entirely in hardware, with one 32-bit word per entry. When a process starts, the page table is copied to the hardware from memory, at one word every 100 nsec. If each process runs for 100 msec (including the time to load the page table), what fraction of the CPU time is devoted to loading the page tables?
15. Suppose that a machine has 48-bit virtual addresses and 32-bit physical addresses.
   (a) If pages are 4 KB, how many entries are in the page table if it has only a single level? Explain.
   (b) Suppose this same system has a TLB (Translation Lookaside Buffer) with 32 entries. Furthermore, suppose that a program contains instructions that fit into one page and it sequentially reads long integer elements from an array that spans thousands of pages. How effective will the TLB be for this case?
17. Suppose that a machine has 38-bit virtual addresses and 32-bit physical addresses.
   (a) What is the main advantage of a multilevel page table over a single-level one?
   (b) With a two-level page table, 16-KB pages, and 4-byte entries, how many bits should be allocated for the top-level page table field and how many for the next level page table field? Explain.
19. A computer with a 32-bit address uses a two-level page table. Virtual addresses are split into a 9-bit top-level page table field, an 11-bit second-level page table field, and an offset. How large are the pages and how many are there in the address space?